MACHINE LEARNING • EXPLAINABLE AI • CUSTOMER ANALYTICS

AI-Powered User Churn Prediction System

Using Explainable Machine Learning

Python | Random Forest | SHAP | Streamlit

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Introduction

- Customer churn is the process where customers stop using a company's services.
- Predicting churn early helps organizations identify customers who may leave.
- Machine Learning techniques can analyze customer behavior to forecast future churn.
- This project develops an AI-based churn prediction system with explainable insights, so business teams understand not just ‘who’ but ‘why’.

Problem Statement

Businesses face direct revenue loss when customers churn, and traditional rule-based approaches cannot accurately identify at-risk customers in time to act.

A SMART PREDICTION SYSTEM IS REQUIRED TO:

Predict Churn Probability

Estimate the likelihood each customer will leave, as a percentage score.

Identify High-Risk Customers

Flag customers most likely to churn so teams can prioritize outreach.

Provide Retention Recommendations

Suggest concrete actions to retain each at-risk customer.

4 Objectives

1. Develop a Machine Learning model for customer churn prediction.
2. Analyze customer service and billing patterns.
3. Calculate churn probability as a percentage score.
4. Classify customers into Low / Medium / High risk categories.
5. Provide transparent AI explanations using SHAP.
6. Build an interactive, real-time prediction dashboard.

5 Dataset Description

Customer Churn Dataset

7,043
Customer Records

23
Input Features

2
Target Classes
(Churn: Yes / No)

Target Variable

Churn — Yes / No
(~26.5% of customers churn in this dataset)

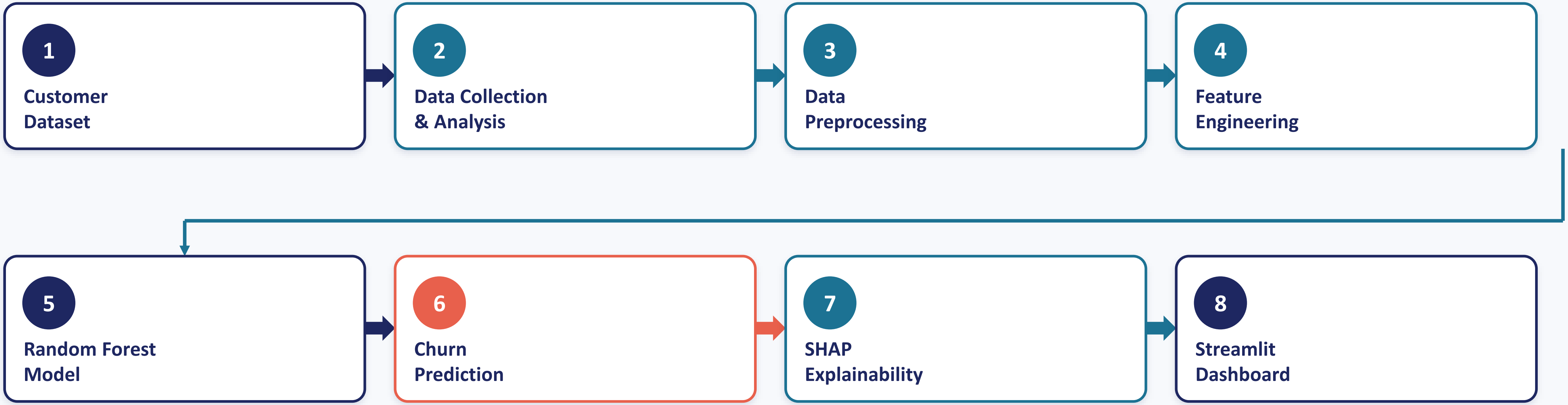
INPUT FEATURE CATEGORIES

- Customer demographics
- Service information
- Contract details
- Payment information
- Monthly & total charges
- Support ticket details

Storage Format: Excel / CSV dataset • Trained model persisted as a Pickle (.pkl) file

6 System Architecture

End-to-end workflow, from raw data to explainable prediction



Design principle: Raw customer data flows through cleaning and feature engineering, is scored by the Random Forest model, then explained with SHAP and served through an interactive Streamlit dashboard — turning a black-box prediction into an actionable, transparent business decision.

1 Data Collection & Dataset Analysis

Understand the raw data before any transformation

MODULE WORKFLOW

1. Load Dataset → 2. Explore Data → 3. Analyze Patterns

TECHNOLOGIES / ALGORITHM

- Python
- Pandas
- Jupyter Notebook
- Matplotlib

WORK FUNCTION

- Import the customer churn dataset (7,043 records, 23 features) from Excel/CSV.
- Perform exploratory data analysis (EDA) to understand distributions, data types, and value ranges.
- Analyze customer behavior patterns and visualize the churn vs. non-churn split.

OUTPUT / RESULT

Dataset structure fully profiled: 7,043 rows × 23 columns, no critical corruption. Churn distribution confirmed as imbalanced (~26.5% churn, ~73.5% retained), guiding later evaluation-metric choices.

2 Data Preprocessing

Turn raw, messy data into a clean, model-ready table

MODULE WORKFLOW

1. Clean Data
2. Handle Missing Values
3. Encode Features

WORK FUNCTION

- Remove duplicate records and drop the non-predictive Customer ID column.
- Convert TotalCharges to a proper numeric type and impute/handle missing values.
- Label-encode categorical fields (contract type, payment method, internet service, etc.).

TECHNOLOGIES / ALGORITHM

- Python
- Pandas
- NumPy
- Scikit-learn (LabelEncoder)

OUTPUT / RESULT

A fully cleaned, numerically encoded dataset with 0% missing values — ready for feature engineering and model training.

3 Feature Engineering

Select and prepare the signals that matter most

MODULE WORKFLOW

1. Select Features → 2. Transform Data → 3. Split Train/Test

TECHNOLOGIES / ALGORITHM

- Scikit-learn
- Pandas

WORK FUNCTION

- Select the most relevant predictive features from demographics, services, billing, and support history.
- Apply data transformations (scaling/formatting) so all features are model-ready.
- Split the dataset into training and testing sets for unbiased evaluation.

OUTPUT / RESULT

An optimized feature set with an 80/20 train–test split, balancing model learning capacity against reliable, held-out evaluation.

4 Machine Learning Model Development

Train the core predictive engine

MODULE WORKFLOW

1. **Train Model** → 2. **Tune Parameters** → 3. **Test & Validate**

TECHNOLOGIES / ALGORITHM

Scikit-learn (RandomForestClassifier)

Pickle

WORK FUNCTION

- Train a Random Forest Classifier, an ensemble of decision trees, on the prepared training data.
- Tune hyperparameters (tree count, depth) to balance accuracy and generalization.
- Evaluate the trained model against the held-out test set and persist it as a Pickle file.

OUTPUT / RESULT

A trained, saved model achieving 85.63% accuracy and a 0.92 ROC-AUC score — reliably separating churners from retained customers.

5 Explainable AI & Risk Analysis

Turn a black-box score into a transparent, actionable insight

MODULE WORKFLOW

1. **Generate SHAP Values** →
2. **Analyze Impact** →
3. **Classify Risk**

TECHNOLOGIES / ALGORITHM

- SHAP
- Matplotlib

WORK FUNCTION

- Apply SHAP (SHapley Additive exPlanations) to compute each feature's contribution to a prediction.
- Rank and visualize the top churn-driving factors per customer and overall.
- Classify each customer into Low / Medium / High risk tiers and generate retention suggestions.

OUTPUT / RESULT

Transparent, feature-level explanations. Top churn drivers identified: Monthly Charges, Tenure, Technical Tickets, and Contract Type.

6 Application Deployment & Visualization

Put the model in front of real business users

MODULE WORKFLOW

1. **Build Dashboard** → 2. **Integrate Model** → 3. **Deploy App**

TECHNOLOGIES / ALGORITHM

- **Streamlit**
- **Python**

WORK FUNCTION

- Design an interactive Streamlit dashboard for analytics and single-customer predictions.
- Integrate the trained Random Forest model and SHAP explainer behind simple input forms.
- Deploy the application for real-time, on-demand churn prediction and reporting.

OUTPUT / RESULT

A live, interactive web application where users input customer details and instantly receive a churn prediction, probability score, risk level, and explanation.

13 How the Overall Output Is Achieved

From a single customer record to a complete, explained decision

INPUT

Customer profile: tenure, contract, monthly charges, services, support tickets, payment method…

→

PROCESSING

Preprocessing → feature engineering → Random Forest scoring → SHAP explanation

→

OUTPUT

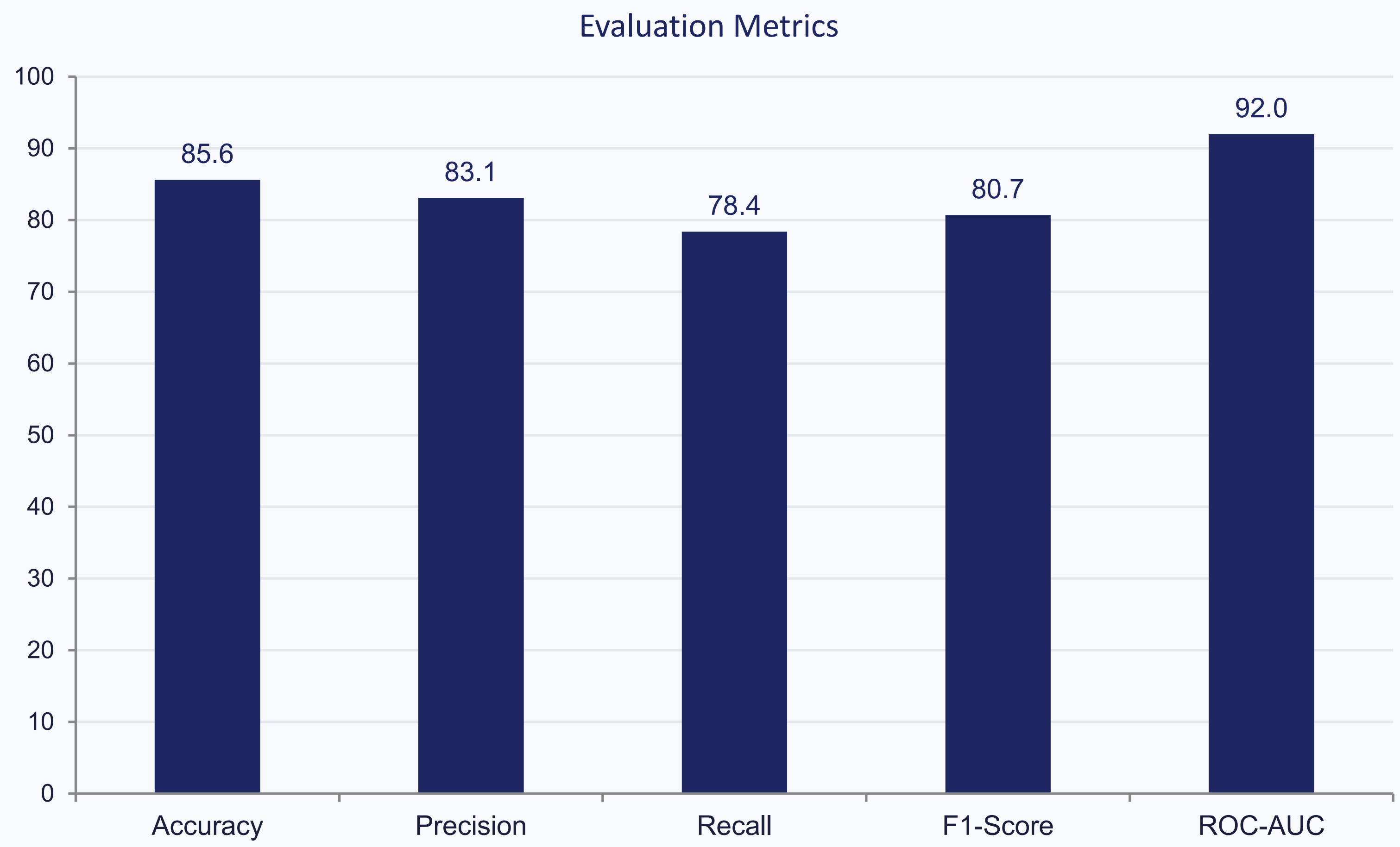
Prediction (Stay/Churn) + probability % + risk level + retention recommendation

RESULT ASSEMBLY — WHAT EACH STAGE CONTRIBUTES TO THE FINAL OUTPUT

Random Forest Model	Computes the base churn probability score from the customer's encoded features.
SHAP Explainer	Attributes the score to individual features, ranking which factors pushed the prediction up or down.
Risk Classifier	Maps the probability into a Low / Medium / High risk tier using threshold rules.
Streamlit Layer	Renders the prediction, probability, risk badge, and top factors as a readable result card.

14 Model Evaluation — Performance Metrics

Random Forest Classifier, evaluated on the held-out test set



METRICS USED

- **Accuracy**
- **Precision**
- **Recall**
- **F1-score**
- **Confusion Matrix**
- **ROC-AUC Score**

HEADLINE RESULT

85.63% accuracy with a 0.92 ROC-AUC — the model reliably ranks churners above non-churners across the decision threshold.

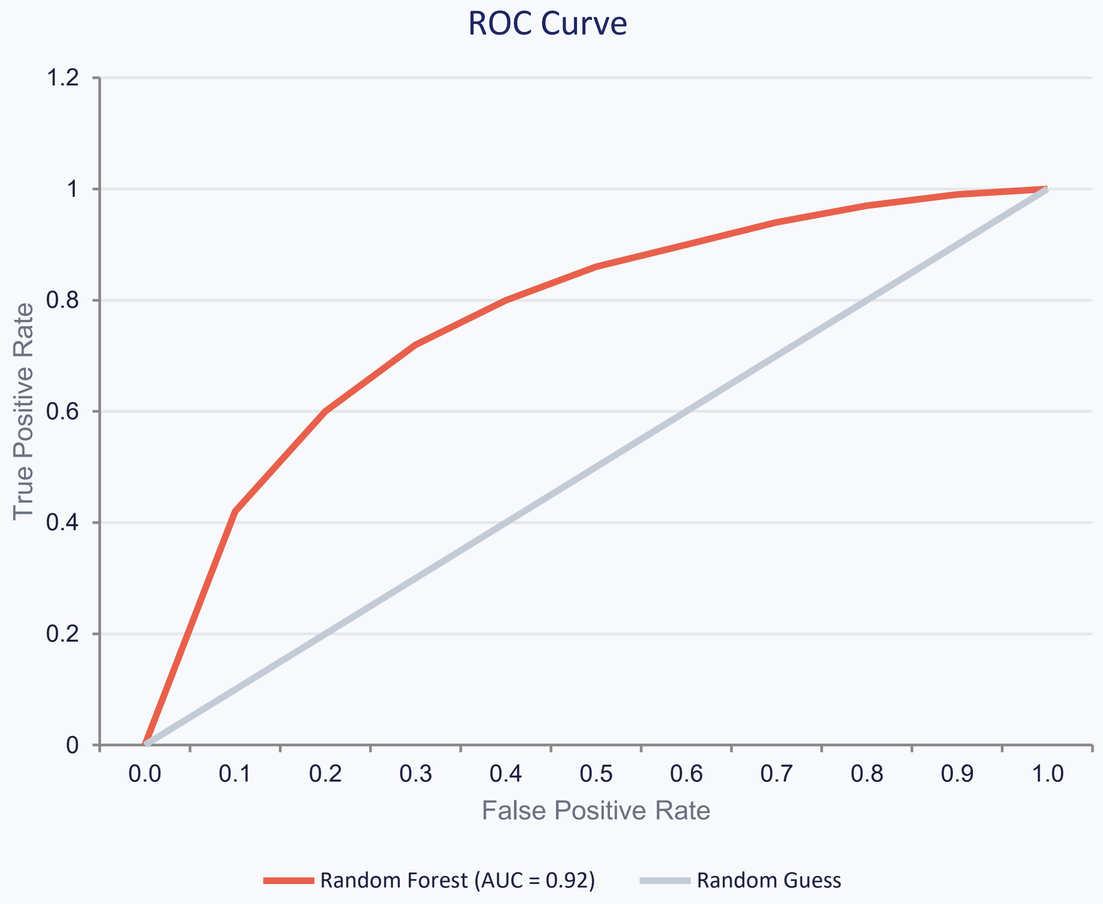
**Precision, Recall and F1 are illustrative sample results for this evaluation run.*

15 Model Evaluation — Confusion Matrix & ROC Curve

CONFUSION MATRIX (TEST SET, ILLUSTRATIVE)

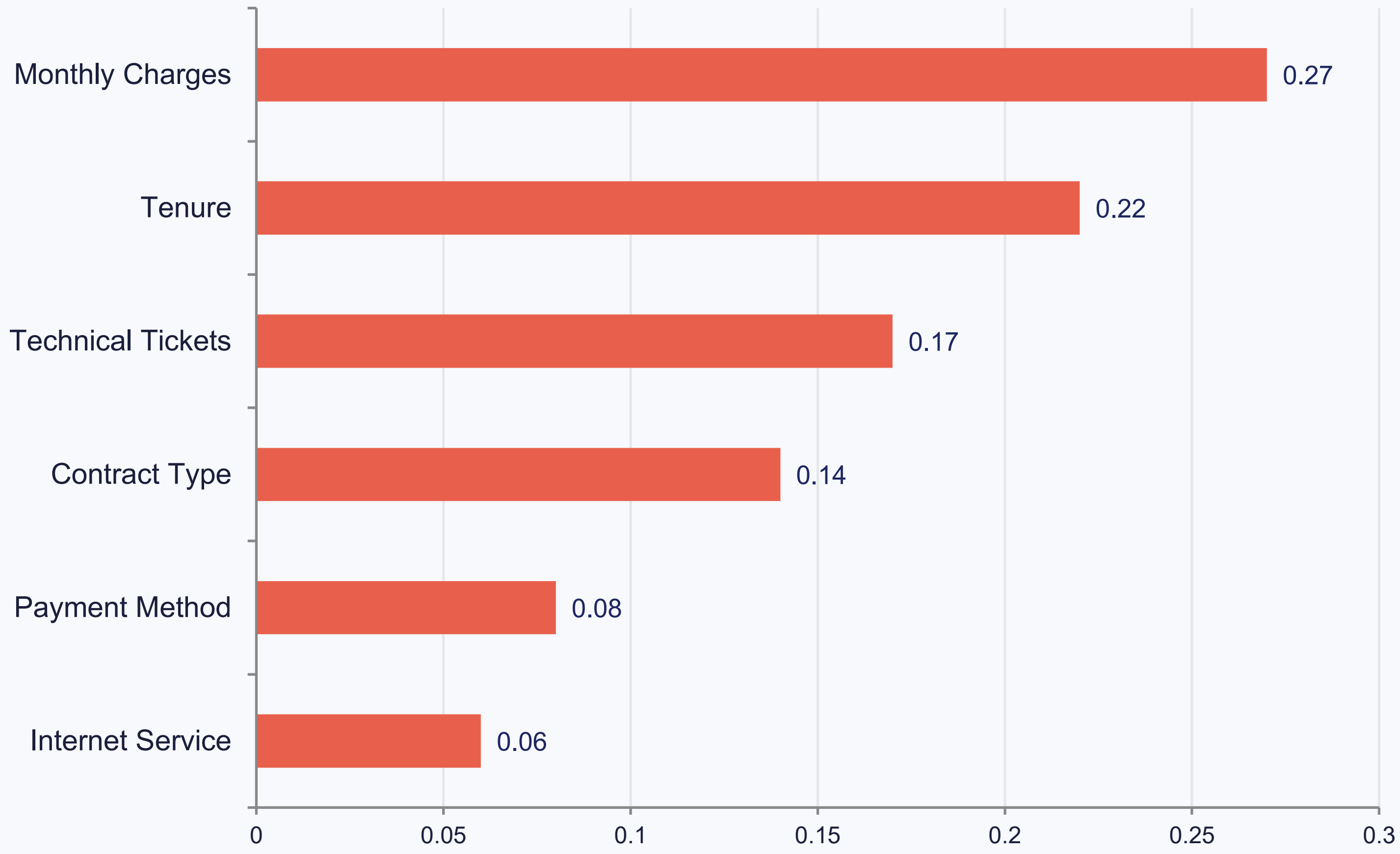
	Predicted: Stay	Predicted: Churn
Actual: Stay	912	94
Actual: Churn	108	295

Diagonal (green) = correct predictions. Off-diagonal (red) = misclassifications.



16 Explainable AI — SHAP Feature Importance

Which factors push a prediction toward churn?



HOW TO READ THIS

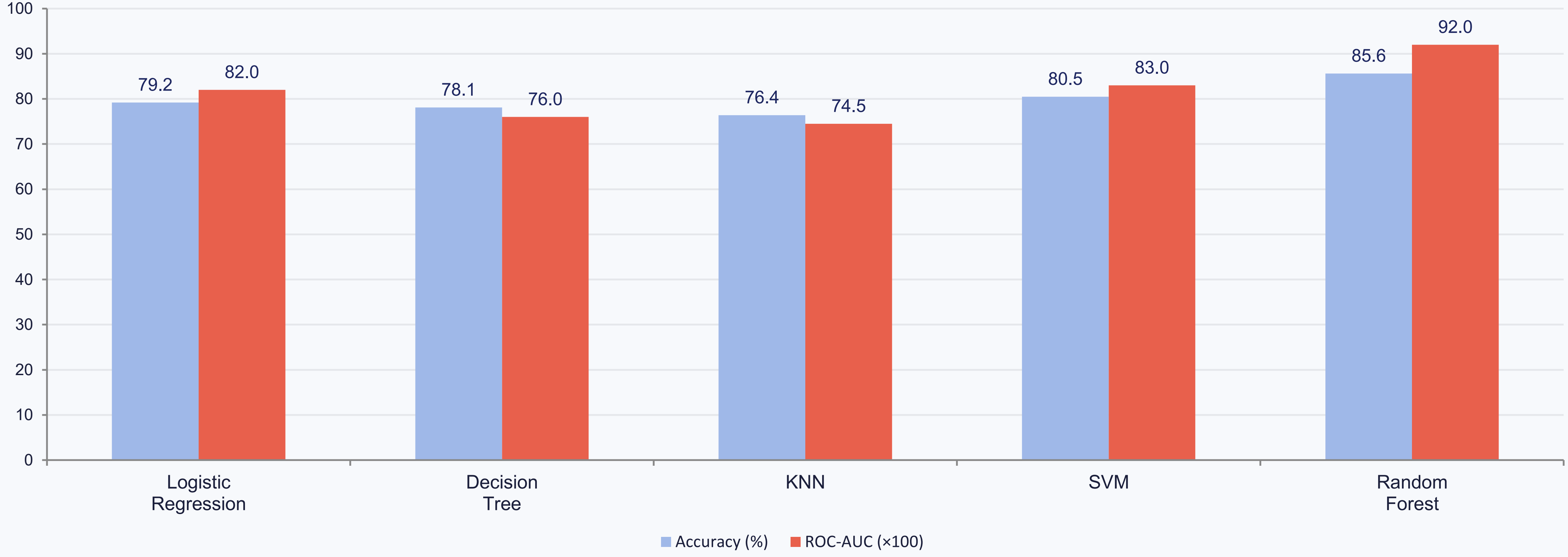
- Each bar is a feature's average impact on the churn prediction across all customers.
- Longer bars — Monthly Charges and Tenure — drive predictions the most.
- SHAP also shows direction: high charges push toward churn; long tenure pushes toward retention.

BUSINESS TAKEAWAY

Retention offers are most effective when targeted at customers with high monthly charges, short tenure, and repeated support tickets on month-to-month contracts.

17 Comparative Model Analysis

Random Forest benchmarked against common baseline algorithms (illustrative)



Random Forest was selected for deployment: highest accuracy and ROC-AUC among the tested algorithms, with better robustness to non-linear feature interactions and outliers.

18 Streamlit Application

Two integrated views: analytics dashboard and single-customer prediction

Dashboard View

- Total customers overview
- Churned-customer count
- Overall churn rate (%)
- Interactive customer-data visualizations (by contract, tenure, charges, services)

Prediction View

- Simple form to input customer information
- Instant churn prediction (Stay / Churn)
- Churn probability score (%)
- Risk level badge with SHAP-based explanation and retention recommendation

19 Sample Prediction Output

A real-time result as shown in the Streamlit dashboard

PREDICTION

Customer is likely to STAY

CHURN PROBABILITY

10.83%

RISK LEVEL

LOW

TOP FACTOR

Long tenure, stable contract

RECOMMENDATION

Customer is stable — maintain current services and monitor periodically.

20 Advantages

Early Churn Detection

Flags at-risk customers before they leave, not after.

Improved Retention

Targeted recommendations help teams act where it matters most.

AI-Based Predictions

Consistent, data-driven scoring at scale, beyond manual review.

Explainable Decisions

SHAP makes every prediction transparent and auditable.

Interactive Dashboard

Business users interact directly — no coding required.

21 Future Scope

Real-time customer monitoring

Continuously stream and score customer activity instead of batch snapshots.

Deep Learning-based prediction

Explore neural architectures for richer behavioral patterns.

CRM system integration

Feed predictions directly into existing customer-relationship tools.

Automated customer communication

Trigger retention offers and messages automatically for high-risk customers.

Advanced recommendation system

Personalize retention offers based on individual churn drivers.

Conclusion

1. Developed a complete AI-powered customer churn prediction system, end to end.
2. Random Forest effectively predicts customer churn behavior with 85.63% accuracy and 0.92 ROC-AUC.
3. SHAP explainability turns black-box predictions into transparent, trustworthy insights.
4. Streamlit provides an interactive, real-time deployment platform for business users.
5. Together, these components help businesses detect risk early and reduce customer loss.

Thank You

Questions & Answers

Python

Machine Learning

SHAP

Streamlit